

HemoScan AI - Clinical Hematology Report

Hematology Clinical Evaluation Report

Patient Identification

* Name: ASDFGHKOPL[

* Phone: 9468213961

* Age: 19.0 Years

* Gender: Male

Clinical Data Comparison

The following table compares the patient's Complete Blood Count (CBC) results with standard clinical reference ranges for an adult male.

Parameter	Patient Value	Reference Range (Adult Male)	Status
Hemoglobin (Hb)	12.1 g/dL	13.5 – 17.5 g/dL	Low
RBC Count	4.91 x10 ¹² /L	4.50 – 5.90 x10 ¹² /L	Normal
PCV (Hematocrit)	33.5 %	41.0 – 50.0 %	Low
MCV	80.4 fL	80.0 – 100.0 fL	Low-Normal
MCH	28.9 pg	27.0 – 33.0 pg	Normal
MCHC	38.0 g/dL	32.0 – 36.0 g/dL	High

Clinical Interpretation

Upon detailed analysis of the hematological profile, there is a clinical discrepancy between the ML Model prediction and the established diagnostic criteria for anemia in adult males.

- Anemic Status:** According to World Health Organization (WHO) guidelines, anemia in adult males is defined as a Hemoglobin (Hb) level below 13.0 g/dL. With a recorded Hb of 12.1 g/dL and a Packed Cell Volume (PCV) of 33.5%, the patient clinically presents with Mild Anemia. The ML model's "Not Anemic" prediction carries an extremely low confidence score (7.28%), which suggests the model is flagging the results as borderline or inconsistent with its training data.
- Red Cell Indices:** The MCV (80.4 fL) is at the lowest threshold of the normal range, suggesting a trend toward microcytosis (small red blood cells). However, the MCH remains within normal limits.
- MCHC Elevation:** The MCHC (38.0 g/dL) is notably elevated. In clinical practice, a high MCHC (hyperchromia) is relatively rare. It can be indicative of hereditary spherocytosis, or more commonly, it may result from pre-analytical factors such as cold agglutinins, lipemia, or hemolysis within the sample.
- RBC Count:** The RBC count is preserved (4.91), which, when paired with a low Hb, can sometimes be seen in conditions like Thalassemia trait or early-stage iron deficiency, though the indices here are not yet definitive for those pathologies.

Risk Level

Level: Moderate

While the anemia is currently "mild" in terms of hemoglobin concentration, the discrepancy in MCHC and the patient's young age (19) warrant a proactive investigation to identify the underlying etiology and prevent further decline in hematological health.

Possible Anemia Type

Based on the indices provided, the patient presents with:

- * Normocytic Normochromic Anemia (Borderline Microcytic): This is often the earliest stage of nutritional deficiency or may be secondary to an underlying inflammatory process (Anemia of Chronic Disease).
- * Potential Spherocytosis: Suggested by the elevated MCHC, though this requires morphological confirmation.

Suggested Diagnostic Tests

To reach a definitive diagnosis and formulate a treatment plan, the following investigations are recommended:

1. Peripheral Blood Smear (PBS) Examination: Essential to evaluate RBC morphology (looking for spherocytes, target cells, or hypochromia) and to rule out laboratory artifacts causing the high MCHC.
2. Reticulocyte Count: To assess the bone marrow's regenerative response to the low hemoglobin levels.
3. Iron Studies: Including Serum Iron, Ferritin, and Total Iron Binding Capacity (TIBC) to rule out early-stage Iron Deficiency Anemia.
4. Serum Vitamin B12 and Folate Levels: To ensure nutritional adequacy for erythropoiesis.
5. Repeat CBC with Automated and Manual Differential: To confirm the MCHC value and ensure no cold agglutinins are affecting the results.